



RKDF UNIVERSITY, RANCHI

PG Diploma in Computer Application

SCHEME

First Semester

S.No.	Subject Code	Subject Name	Marks Distribution				
			Internal	External		Total	
			Max	Max	Min	Max	Min
1	PGDCA101	Fundamentals of Computer and Information Technology	30	70	21	100	35
2	PGDCA102	PC- Packages and Computerized Accounting System	30	70	21	100	35
3	PGDCA103	Data Communication and Computer Network	30	70	21	100	35
4	PGDCA104	Programming using 'C' & C++	30	70	21	100	35
5	PGDCA105	System Analysis & Design	30	70	21	100	35
Practical							
Total							
			Max		Min		
1	PGDCA151	PC Package and Tally ERP Lab	50		25		
2	PGDCA152	C,C++ and Oracle Lab	50		25		
3	PGDCA153	Project	50		25		



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Choice Based Credit System

First Semester

S.No.	Subject Code	SUBJECT NAME	PERIODS			Credit
			L	T	P	
1	PGDCA101	Fundamentals of Computer and Information Technology	3	0	0	3
2	PGDCA102	PC- Packages and Computerized Accounting System	3	0	0	3
3	PGDCA103	Data Communication and Computer Network	4	0	0	4
4	PGDCA104	Programming using 'C' & C++	4	0	0	4
5	PGDCA105	System Analysis & Design	3	0	0	3
Practicals						
1	PGDCA151	PC Package and Tally ERP Lab	0	0	4	2
2	PGDCA152	C,C++ and Oracle Lab	0	0	4	2
3	PGDCA153	Project	0	0	4	2
Total			17	0	12	23



R.K.D.F. UNIVERSITY, RANCHI

PG Diploma in Computer Application

Semester – I

Course Content

Branch	Subject Title	Subject Code
PGDCA	FUNDAMENTALS OF COMPUTER & INFORMATION TECHNOLOGY	PGDCA101

Unit-1

Introduction to Computer and Information Technology: Brief history of development of computer & generations of computer, Computer system characteristics. Capabilities and limitations block diagram of computer. Types of computer-Analog, Hybrid, digital, micro, mini, mainframe, super computer. Personal computer, types of PCs desktop, laptop, notebook, palmtop etc. Number system Data representation in computers, Number system of computers binary, octal, hexadecimal, representation & their conversion, Coding system ASCII, BCD, EDCDIC etc.

Unit-2

INPUT/OUTPUT devices: keyboard, mouse, monitor, trackball, joystick, digitizing table, scanners, digital cameras, MICR, OCR, OMR, Bar-code reader, Voice recognition, light pen, touch screen, devices, printer, plotter.

Unit-3

Storage device: Data storage and retrieval methods-sequential, direct and index sequential-various storage devices-magnetic tape, magnetic disks, cartridge tape, data drives hard disk drives, floppy disks, optical disks-CD, VCD, CDR, CDRW, DVD.

Unit-4

Computer software: types of software, system software, application software, operating system, utility program, assemblers, compilers and interpreter. Operating system functions, Types batch, single user, multi user, multiprogramming, multiprocessing, Programming languages, machine, assembly, high level, 4GL, their merits and demerits. Computer virus – types of virus, virus detection & prevention virus on network.

Unit-5

Data Communication & networks: analog and digital signals, modulations, amplitude modular (am), frequency modulation (fm), phase modulation (pm), communication process, direction of transmission flow, simplex, half duplex, full duplex. Types of network LAN, WAN, MAN etc, Topologies of LAN ring, bus star, mesh and tree topologies, communication protocols TCP/IP protocol suit. Communication channels media twisted, coaxial fiber optic, serial and parallel communication, Network operating system (NOS), bridges, hub, routers, repeater and gateways. Modem working and characteristics. Types of connections- dialup leased lines, ISDN, broadband.

Text & Reference Books:

01. Computer fundamentals, P.K. Sinha, BPB
02. Computer today by S.K. Basandra Galgotia Publications.
03. Fundamentals of information by Alexos Leon & Mathews Leon, Vikas Publishing House, New Delhi.



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Semester – I

Course Content

Branch	Subject Title	Subject Code
PGDCA	PC PACKAGES & COMPUTERIZED ACCOUNTING SYSTEM	PGDCA102

Unit-1

Fundamental of DOS & Windows: Fundamental of DOS booting process, internal and external commands, creating and executing batch files and directories creating text files. Introduction to windows features, various versions of windows, origin of windows parts of windows screen types and anatomy of windows using.

Unit-2

Introduction to word processing (MS-word): Advantages of word processing, editing a file using paragraphs, bullets, indentation, ect. Formatting features, printing the documents, it includes paper-size, margins, header and footer, page no., using macros. Advance word processing, header and footers. Finding text, mail merge and other application, mathematical calculations, table handing.

Unit-3

Introduction to spread sheet (MS-Excel): Definition and advantages of electronic worksheet, working of spread sheet, range and related operations. Setting saving and retrieving work sheet file, inserting deleting coping & moving of data cells, inserting and deleting rows & columns, protecting cell printing a worksheet, erasing a worksheet, graphs, creation, types of graphs creating a chart sheet 3D column charts, moving and changing the size of chart, printing the chart.

Unit-4

Introduction to Powerpoint (MS- Powerpoint): Creating a presentation, inserting/deleting slides, different slide views, editing slides,. Slide transition & editing special effects inserting sound, picture, chart, organization chart.

Unit-5

Accounting software Tally ERP 9: Basic principles of double entry accounting system, creating new company security controls, groups, ledger, voucher type, modifying, new company, voucher entry, generating profit & loss account, trial balance and balance sheet, backup & restore.

Text & Reference Books:

01. Comdex Computer Course Kit (Windows 7 with office 2010), Gupta vikas, Dreamtech Publication.
02. Mastering MS Office 2000, Professional Edition by Courter, BPB Publication.
03. MS Office 2000 Training Guide by Maria, BPB Publication.
04. PC Software, Ravi Taxalli, BPB
05. Computer Fundamental by P.K. Sinha
06. Financial Accounting with Tally 9.001 edition by Vikas Gupta.
07. Mastering Tally ... ERP 9 By A.K. Nandhani



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Semester – I

Course Content

Branch	Subject Title	Subject Code
PGDCA	DATA COMMUNICATION & COMPUTER NETWORK	PGDCA103

Unit-1

Introduction to Data Communication – Network models, protocols and architecture, standards organizations, line configuration, topology, transmission mode, classification of networks, OSI reference model, TCP/IP model.

Unit-2

Analog and digital signals, Data encoding, parallel and serial transmission, modems, transmission media: guided media, unguided media, transmission impairment, performance, Synchronous and asynchronous transmission.

Unit-3

Multiplexing, LLC, error detection and correction, flow control, HDLC, LANs- applications, architecture, Ethernet, 802.3 LANs, token ring, FDDI, IEEE 802.6, circuit switching, packet switching, message switching, connection oriented and connectionless services.

Unit-4

Principles of internetworking– connection– oriented, connectionless, Routing concepts, routing algorithms– distance-vector routing, link state routing, shortest path routing. Congestion control, QOS, internetworking, network devices.

Unit-5

Network security requirements and attacks, public key and private key encryption and digital signatures, digital certificate, firewalls, IDS (Intrusion Detection System)

Text & Reference Books:

01. Computer networks– A.S. Tanenbaum. PHI
02. Data communication and networking – Behrouz A. Forouzan. TM



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Semester – I

Course Content

Branch	Subject Title	Subject Code
PGDCA	SYSTEM ANALYSIS AND DESIGN	PGDCA104

Unit-1

The system concept: characteristics, elements and types of a system, the system development life cycle, considerations, for candidate systems prototyping. The role of system analyst.

Unit-2

System planning and initial investigation: Information Gathering, information gathering tools. Structured analysis, the tools of structured analysis (DFD, Data Dictionary, Decision tree and Pseudo codes Decision Tables), PROS and CONS of each tool, system performance definition description of outputs, feasibility study. Cost/ Benefit analysis, Data analysis, Cost/ Benefit analysis, the system proposal.

Unit-3

Stages of system design: Design methodologies, development activities, input design, output design forms design, types of forms, basics of form design layout considerations and forms control.

Unit-4

File structure: File organization, objectives of database, data structure, system testing and quality assurance, why system testing, what do we test for, the test plan quality assurance, trends in testing, role of data processing auditor, training and documentation.

Unit-5

Implementing and software maintenance: conversion combating resistance to change, post implementation review, software maintenance, hardware/software selection and the computer contract, suppliers, procedure for hardware/software selection, financial considerations in selection, the computer contract system security disaster recovery planning.

Text & Reference Books:

01. System analysis and design, Elias M. Awad, Galgotia Publication (P) Ltd.
02. System analysis and design, International Ed. Perry Edwards, McGraw Hill Pub.



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Semester – I

Course Content

Branch	Subject Title	Subject Code
PGDCA	PROGRAMMING IN C & C++	PGDCA105

Unit-1

Introduction to “C” Language: Fundamentals, simple I/O statements, reading and writing, data types constants, variable, operators & expressions, library function, control statements, if-else, while, do-while, goto, for statements switch, break, looping statements, functions recursion, arrays, multidimensional arrays, strings & pointers.

Unit-2

Programming in C++, functions, class, object, constructor and destructor: Call by reference, call by value, return by reference, inline function, constant argument, function overloading, static member function, static data member,. Classes: implementing class, classes and members, accessing class members, implementing class methods, array of object, friend function. Constructor & destructors: parameterized constructor, multiple constructor, constructor with default argument, copy constructor, destructor.

Unit-3

Operator overloading & type casting: Operator overloading, unary operator overloading, binary operator overloading, manipulates string using operator overloading, type conversions: basic to class, class to basic, class to class.

Unit-4

Inheritance, virtual function: single inheritance, multilevel inheritance, multiple inheritance, hybrid inheritance, hierarchical inheritance, virtual base class, abstract class.

Unit-5

Pointer & File: Pointer to object, this pointer, virtual function and pure virtual function. File: opening and close file, detecting end of the file .

Text Books:

01. Let us C by Yaswant Kanetkar BPB
02. Object oriented Programming with C++, E.Blagurusamy, Tata mc Graw-Hill
03. C++ Complete reference, Herbert Schildt, TMH.
04. ANSI C programming, E.Blagurusamy, TMH



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SCHEME

Second Semester

S.No.	Subject Code	Subject Name	Marks Distribution				
			Internal	External		Total	
			Max	Max	Min	Max	Min
1	PGDCA201	Programming in Java	30	70	21	100	35
2	PGDCA202	Web Technologies	30	70	21	100	35
3	PGDCA203	Software Engineering	30	70	21	100	35
4	PGDCA204	Data Structures	30	70	21	100	35
5	PGDCA205	Operating System	30	70	21	100	35
Practical							
			Total				
			Max		Min		
1	PGDCA251	Programming in Java Lab	50		25		
2	PGDCA252	Web Technologies Lab	50		25		
3	PGDCA253	Data Structures Lab	50		25		



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Choice Based Credit System

Second Semester

S.No.	Subject Code	SUBJECT NAME	PERIODS			Credit
			L	T	P	
1	PGDCA201	Programming in Java	4	0	0	4
2	PGDCA202	Web Technologies	3	0	0	3
3	PGDCA203	Software Engineering	3	0	0	3
4	PGDCA204	Data Structures	3	0	0	3
5	PGDCA205	Operating System	4	0	0	4
Practicals						
1	PGDCA251	Programming in Java Lab	0	0	4	2
2	PGDCA252	Web Technologies Lab	0	0	4	2
3	PGDCA253	Data Structures Lab	0	0	4	2
Total			17	0	12	23



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Semester – II

Course Content

Branch	Subject Title	Subject Code
PGDCA	Programming in Java	PGDCA201

Unit - I

OOPs concepts: Basic Concepts of Object-Oriented Programming (Objects and Classes, Data abstraction and encapsulation, Inheritance, Polymorphism, Dynamic binding, Message communication), difference between procedure oriented and object oriented approach, Benefits of OOP's; Applications of OOP's, Object-Oriented languages.

Object oriented programming with JAVA: Byte code, Java virtual machine, Java Development Kit, java tokens, constants, variables, data types, operators, expressions, control structures, defining class, creating objects, accessing class members, method overloading, static members

Unit – II

Inheritance: Defining a subclass, subclass constructor, multilevel inheritance, Hierarchical inheritance. Overriding methods, Final variables, methods, and classes, Abstract Methods and Classes.

Visibility Control: Public access, friendly access, protected access, private access, private protected access.

Arrays: One dimensional array, declaration, creation and initialization of arrays, Array length, Two dimensional array

Unit – III

Strings: String arrays, String methods, StringBuffer class

Interfaces: Defining interfaces, Extending Interfaces, Implementing Interfaces. Accessing Interface variables

Packages: Java API packages, Defining a package, Creating and Accessing packages, Adding class to a package, Hiding Classes.

Multithreaded Programming: Creating Thread, Extending the Thread class, Stopping and Blocking a Thread, Life cycle of a Thread.

Unit - IV

Errors and Exception Handling: Fundamentals, error types, exception types, using Try and catch, finally statement, Built-in exceptions.

Applet Programming: Local and remote applets, Applet Life Cycle, Creating an executable Applet, Applet tag, Adding Applet to a HTML file, Passing parameters to Applets

References:

1. Balaguruswamy, E. Fundamentals of Java
2. Daniel Dang An Introduction to Java Programming, PHI, New Delhi
3. Deitel & Dietel JAVA, How to Program, Pearson Education Asia
4. Liang An Introduction to Java Programming PHI



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Semester – II

Course Content

Branch	Subject Title	Subject Code
PGDCA	Web Technologies	PGDCA202

Unit-1

Internet Basics: Basic concepts, communicating on the internet, internet domains, internet server identities, establishing connectivity on the internet client IP address, How IP addressing came into existence? A brief overview TCP/IP and its services, transmission control protocol.

Unit-2

Introduction To HTML: Information Files Creation, Web Server, Web Client/Browser, Hyper Text Markup Language (HTML Tags, Paired Tags, Singular Tags), Commonly Used HTML Commands (Document Head, Document Body), Title and Footer, Text Formatting (Paragraph Breaks, Line Breaks), Emphasizing Material in a Web Page (Heading Styles, Drawing Lines).

Unit-3

Basic Formatting Tags: HTML Basic Tags, Text Formatting (Paragraph Breaks, Line Breaks), Emphasizing Material in a Web Page (Heading Styles, Drawing Lines), Text Styles (Bold, Italics, Underline), Other Text Effects (Centering (Text, Images etc.), Spacing (Indenting Text), HTML Color Coding.

Unit-4

Lists Type of Lists (Unordered List (Bullets), Ordered Lists (Numbering), Definition Lists.

Unit-5

Adding Graphics To HTML Documents: Using The Border Attribute, Using The Width And Height Attribute, Using The Align Attribute, Using The Alt Attribute.

Unit-6

Tables: Introduction (Header, Data rows, The Caption Tag), Using the Width and Border Attribute, Using the Cell padding Attribute, Using the Cell spacing Attribute, Using the BGCOLOR Attribute, Using the COLSPAN and ROWSPAN Attributes Tag.

Unit-7

Linking Documents: Links (External Document References, Internal Document References), Image As Hyperlinks.

Unit-8

Frames: Introduction to Frames: The tag, The tag, Targeting Named Frames. DHTML: Cascading Style Sheets, Style

Unit-9

Introduction to JavaScript: Introduction to JavaScript: JavaScript in Web Pages (Netscape and JavaScript, Database Connectivity, Client side JavaScript, Capturing User Input); The Advantages of JavaScript (an Interpreted Language, Embedded within HTML, Minimal Syntax -Easy to Learn, Quick Development, Designed for Simple, Small Programs, Performance, Procedural Capabilities, Designed for Programming User Events, Easy Debugging and Testing, Platform Independence/Architecture Neutral); Writing JavaScript into HTML.

Unit-10

Forms Used by a Web Site: The Form Object, The Form Object's Methods (The Text Element, The Password Element, The Button Element, The Submit (Button) Element, The Reset (Button) Element, The Checkbox Element, The Radio Element, The Text Area Element, The Select and Option Element, The Multi Choice Select Lists Element) Other Built-In Objects in JavaScript (The String Object, The Math Object, The Date Object), User Defined Objects (Creating a User Defined Object, Instances, Objects within Objects).

Text Books:

1. Internet for Everyone: Alexis Leon, 1st Edition, Leon Techworld, Publication, 2009.
2. Greenlaw R; Heppe, "Fundamentals of Internet and WWW", 2nd Edition, Tata McGraw-Hill, 2007.
3. Raj Kamal, "Internet & Web Technologies", edition Tata McGraw-Hill Education.2009.
4. Chris Payne, "Asp in 21 Days", 2nd Edition, Sams Publishing, 2003 PDCA.
5. A Beginner's Guide to Html [Http://www.Ncsa.Nine.Edit/General/Internet/WWW/Html.Prmter](http://www.Ncsa.Nine.Edit/General/Internet/WWW/Html.Prmter)

E-Books/ Online learning material:

1. https://www.tutorialspoint.com/html/html_tutorial.pdf
2. <https://www.w3schools.com/js/>
3. <https://www.w3schools.com/html/>
4. https://www.cs.uct.ac.za/mit_notes/web_programming.html
5. http://www.pagetutor.com/table_tutor/index.html



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Semester – II

Course Content

Branch	Subject Title	Subject Code
PGDCA	Software Engineering	PGDCA203

Unit-1

Introduction to the Discipline, The Software Process, Software Engineering Practice, Software Development Myths.

Unit-2

Prescriptive Process Models (The Waterfall Model, Incremental Process Models, Evolutionary Process Models, Concurrent Models), Specialized Process Models (Component-Based Development, The Formal Methods Model, Aspect-Oriented Software Development), The Unified Process, Phases of the Unified Process, Personal and Team Process Models (Personal Software Process, Team Software Process).

Unit-3

Requirements Engineering, Understanding of Software Requirements, Building the Analysis Model, The Design Process, Design Concepts, The Design Model (Data Design Elements, Architectural Design Elements, Interface Design Elements, Component-Level Design Elements, Deployment-Level Design Elements).

Unit-4

Approach to Software Testing, Unit Testing, Integration Testing, Validation Testing, System Testing, Debugging, Software Testing Fundamentals, White-Box Testing, Basis Path Testing, Control Structure Testing, BlackBox Testing.

Unit-5

A Framework for Product Metrics, Metrics for the Requirements Model, Metrics for the Design Model, Metrics in the Process and Project Domains, Software Measurement.

Unit-6

Software Maintenance, Reengineering, Software Reengineering, Reverse Engineering, Restructuring, Forward Engineering, The Economics of Reengineering.

Text Books:

1. Software Engineering—A Practitioner's Approach, Roger S. Pressman and Bruce R. Maxim, Eighth Edition, 2015, McGrawHill.

Reference Books:

1. An Integrated Approach to Software Engineering, Pankaj Jalota, Third Edition, 2005, Narosa Publishing House;
2. Software Engineering, Ian Sommerville, Ninth Edition, 2011, Addison-Wesley.



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Semester – II

Course Content

Branch	Subject Title	Subject Code
PGDCA	Data Structures	PGDCA204

Unit-1

Stack and Queue: contiguous implementations of stack, various operations on stack, various polish notations-infix, prefix, postfix, conversion from one to another-using stack; evaluation of post and prefix expressions. Contiguous implementation of queue: Linear queue, its drawback; circular queue; various operations on queue; linked implementation of stack and queue- operations

Unit-2

General List and Trees: list and it's contiguous implementation, it's drawback; singly linked list-operations on it; doubly linked list-operations on it; circular linked list; linked list using arrays. Tree definitions-height, depth, order, degree, parent and child relationship etc; Binary Trees- various theorems, complete binary tree, almost complete binary tree; Tree traversals-preorder, in order and post order traversals, their recursive and non recursive implementations; expression tree- evaluation; linked representation of binary tree-operations. Threaded binary trees; forests, conversion of forest into tree. Heap-definition

Unit-3

Searching, Hashing and Sorting: requirements of a search algorithm; sequential search, binary search, indexed sequential search, interpolation search; hashing-basics, methods, collision, resolution of collision, chaining; Internal sorting- Bubble sort, selection sort, insertion sort, quick sort, merge sort on linked and contiguous list, shell sort, heap sort, tree sort.

Unit-4

Graphs: related definitions: graph representations- adjacency matrix, adjacency lists, adjacency multilist; traversal schemes- depth first search, breadth first search; Minimum spanning tree; shortest path algorithm; kruskals & dijkstras algorithm.

Text Books :

1. Brijesh Bakariya. Data Structures and Algorithms Implementation through C, BPB Publications.
2. Data Structures, Schaum Series, TMH.
3. Kruse R.L. Data Structures and Program Design in C; PHI
4. Aho Alfred V., Hopperoft John E., Uilman Jeffrey D., “Data Structures and Algorithms”, AddisonWesley

Reference Books:

1. Horowitz & Sawhaney: Fundamentals of Data Structures, Galgotia Publishers.
2. Yashwant Kanetkar, Understanding Pointers in C, BPB Publications.
3. Horowitz, S. Sahni, and S. Rajasekaran, Computer Algorithms, Galgotia Pub. Pvt. Ltd., 1998.



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Semester – II

Course Content

Branch	Subject Title	Subject Code
PGDCA	Operating System	PGDCA205

Unit-1

Fundamentals of Operating system: What is Operating system? Functions of an operating system. Operating system as a resource manager. Structure of operating system (Role of kernel and Shell). Views of operating system. Evolution and types of operating systems.

Unit-2

Process management: Definition of process, process states, Process Control Block, Scheduling Queues, Schedulers, context switch.

Unit-3

Inter Process Communication: Communication/message passing mechanisms, threading, multithreading models, multicore programming, Fundamental concepts of OpenMP.

Unit-4

Process Synchronization: Cooperating process, critical section problem, mutex locks, semaphores, deadlock and starvation, bounded buffer problem, reader-writer problem.

Unit-5

CPU scheduling: Basic concepts, Scheduling criteria, single processor scheduling, multiprocessor scheduling, real time scheduling, Algorithm Evaluation. Deadlock: Definition, necessary conditions, Resource Allocation Graph, Prevention, Avoidance, Detection and Recovery.

Unit-6

Memory Management: Address binding, Dynamic linking and loading, Contiguous memory allocation techniques (fixed and variable sized partitions), Fragmentation and its types, Non-Contiguous memory allocation techniques, Paging, Segmentation, paging with segmentation, Need of Virtual memories, Demand paging, performance measuring of demand paging, Page replacement Algorithms, allocation of frames, Concept of Thrashing.

Unit-7

Device Management: Secondary storage structure, disk scheduling, Disk management, RAID structure, Role of I/O traffic controller, scheduler.

Unit-8

File Management: File concepts, access methods, directory and disk structure, file system structure, file system and directory implementation, Protection and Security. Case Studies: LINUX Operating System and Windows Operating System.

Text Books:

1. Operating System Principles by Abraham Silberschatz and Peter Baer Galvin, Seventh Edition, Published by Wiley-India.
2. Operating Systems by Stuart E. Madnick, John J. Donovan, Published by MacGraw-Hill.

Reference Books:

1. Principals of Operating System by Naresh Chauhan, Published by OXFORD University Press, India.
2. Operating Systems by Sibsankar Haldar and Alex A. Aravind, Published by Pearson Education.
3. Operating system by Stalling, W., Sixth Edition, Published by Prentice Hall (India)